

Institute of Information Technology

Jahangirnagar University, Savar, Dhaka

Subject: Principles of Economics | Tutorial Answer Key

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Tutorial Set: Set 1

Total Marks: 30

Chapters: 1, 2 & 4

Section A: MCQ Answers

10 Marks

Q	Ans	Explanation
Q1	b	Scarcity means society's resources are limited but human wants are unlimited, forcing choices.
Q2	c	Opportunity cost is the value of the best alternative given up. Going to university means giving up income, leisure, and work experience.
Q3	c	Law of Demand: price and quantity demanded move in opposite directions (inverse relationship).
Q4	b	A change in consumer income shifts the entire demand curve. Only a change in the price of the good itself causes a movement along the curve.
Q5	c	A point inside the PPF means the economy is not using all its available resources — it is operating inefficiently.
Q6	b	When price is below equilibrium, quantity demanded exceeds quantity supplied, creating a shortage (excess demand).
Q7	c	Tea and sugar are used together. If the price of tea rises, demand for sugar falls. This is the definition of complementary goods.
Q8	c	'Higher cigarette taxes reduce smoking' is a factual, testable claim (positive statement). The other options involve value judgments (normative).
Q9	b	Principle 10 describes the short-run inverse relationship between inflation and unemployment, as shown by the Phillips Curve.
Q10	c	Set $Q_d = Q_s$: $20 - 3P = 10 + 2P \Rightarrow 10 = 5P \Rightarrow P^* = 2$. This is the equilibrium price.

Section B: Short Answer Answers

12 Marks

1. Explain the concept of 'Opportunity Cost' with a Bangladesh example. [3]

Definition: Opportunity cost is the value of the next best alternative that is given up when a decision is made. It is the real cost of any choice.

Key points:

- Every decision involves a trade-off. To get one thing, you must give up something else.
- Opportunity cost includes both explicit costs (money paid) and implicit costs (hidden, foregone alternatives).
- Even a 'free' choice has an opportunity cost.

Bangladesh Example: A garments worker in Dhaka earns Tk. 8,000 per month. She decides to leave her job and start a small tailoring business. The opportunity cost of starting her business is Tk. 8,000 per month — the salary she gives up — even though no direct payment is made for this cost. If her business earns less than Tk. 8,000, she has not fully covered her opportunity cost.

2. Distinguish between a 'movement along' a demand curve and a 'shift' of the demand curve. [3]

Movement along the demand curve:

- Caused only by a change in the price of the good itself.
- The demand curve stays in the same position. Only the point on the curve changes.
- Technically called a 'change in quantity demanded'.
- Example: Tea price falls from Tk. 20 to Tk. 15 per cup. Consumers buy more tea. The point moves down along the same demand curve.

Shift of the demand curve:

- Caused by a change in factors other than the price of the good: income, tastes, prices of related goods, expectations, or number of buyers.
- The entire curve moves to a new position (left or right).
- Technically called a 'change in demand'.
- Example: Winter arrives and people want more tea at every price level. The demand curve shifts to the right.

3. What is the Circular-Flow Diagram? Identify its two main groups and two main markets. [3]

Definition: The Circular-Flow Diagram is a visual economic model that shows how money and goods flow between the two main groups in an economy through two types of markets.

Two Main Groups:

- Households: They own the factors of production (labour, land, capital) and buy goods and services from firms.
- Firms: They produce goods and services, and hire factors of production from households.

Two Main Markets:

- Market for Goods and Services: Firms sell goods; households buy them. Money flows from households to firms as spending.
- Market for Factors of Production: Households sell labour, land, and capital; firms buy them. Money flows from firms to households as wages, rent, and profit.

The two loops are: the Inner Loop (flow of real goods and factors) and the Outer Loop (flow of money/dollars).

4. State the Law of Supply. Why does the supply curve slope upward? Use the garment industry as an example. [3]

Law of Supply: Other things being equal, when the price of a good rises, the quantity supplied rises; and when the price falls, the quantity supplied falls. Price and quantity supplied move in the same direction (direct relationship).

Reason for upward slope: When the price of a good is high, producing it becomes more profitable. Sellers are willing to work longer hours, hire more workers, and expand production. When the price is low, production is less profitable, so sellers reduce output.

Bangladesh Garment Example: When the global price of t-shirts rises, Bangladeshi garment factories increase production. They run extra shifts, hire additional workers, and purchase more fabric. Quantity supplied increases. When the price falls, factories cut back production. This upward relationship confirms the Law of Supply.

Section C: Long Answer

8 Marks

1. $Q_d = 50 - 4P$ and $Q_s = 10 + 6P$ (Rice Market) [8]

(a) **Equilibrium Price and Quantity** [2 marks]

At equilibrium: $Q_d = Q_s$

$$50 - 4P = 10 + 6P$$

$$50 - 10 = 6P + 4P$$

$$40 = 10P$$

$$P^* = 4 \text{ (Equilibrium Price)}$$

Substituting $P^* = 4$ into the demand equation:

$$Q^* = 50 - 4(4) = 50 - 16 = 34$$

$$Q^* = 34 \text{ (Equilibrium Quantity)}$$

(b) **Surplus or Shortage at $P = 5$** [2 marks]

At $P = 5$:

$$Q_d = 50 - 4(5) = 50 - 20 = 30$$

$$Q_s = 10 + 6(5) = 10 + 30 = 40$$

Since $Q_s (40) > Q_d (30)$: There is a SURPLUS of 10 units.

Explanation: The price of 5 is above the equilibrium price of 4. Sellers want to supply more than buyers are willing to purchase. Sellers will respond by lowering prices until the market returns to $P^* = 4$.

(c) **Consumer Income Rises — 3-Step Method Analysis** [2 marks]

Step 1 — Which curve shifts? An increase in consumer income affects the demand side. Rice is a normal good, so demand increases. The supply curve is unaffected.

Step 2 — Direction of shift: Demand increases. The demand curve shifts to the RIGHT.

Step 3 — New equilibrium: At the original price of $P^* = 4$, there is now excess demand (shortage). Sellers raise prices in response.

New equilibrium result:

- Equilibrium Price rises (above Tk. 4).
- Equilibrium Quantity also rises.
- Both price and quantity increase when demand increases, with supply unchanged.

(d) Adam Smith's Invisible Hand [2 marks]

Definition: Adam Smith (1776) described the 'Invisible Hand' as the mechanism by which prices in a free market automatically guide the decisions of buyers and sellers to produce outcomes that benefit society as a whole — without any central authority directing them.

How it applies to this scenario: When consumer income rises and demand for rice increases, the market price rises. This higher price acts as a signal to farmers — without any government instruction — to plant more rice. More rice is produced, supply gradually increases, and the price adjusts over time. No planner directed this. The price system (the invisible hand) coordinated the response. This demonstrates Principle 6: Markets are usually a good way to organize economic activity.

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Tutorial Set: Set 2

Total Marks: 30

Chapters: 1, 2 & 4

Section A: MCQ Answers

10 Marks

Q	Ans	Explanation
Q1	c	A fine is a negative incentive — it discourages an action by imposing a penalty (punishment).
Q2	b	Principle 5 states that trade allows countries to specialize in what they do best, making both sides better off.
Q3	c	Microeconomics studies individual decision-makers: households, firms, and specific markets. Macroeconomics studies the economy as a whole.
Q4	b	When both demand and supply increase (both shift right), equilibrium quantity definitely rises. The price effect is ambiguous — it depends on which shift is larger.
Q5	c	An inferior good is one for which demand falls when income rises. Bus rides are replaced by Uber or private transport as income increases.
Q6	c	The PPF shifts outward when the economy gains the ability to produce more — through technological improvement or an increase in available resources.
Q7	c	$ E_p = 1.8 > 1$, so demand is elastic. A 1% rise in price leads to a 1.8% fall in quantity demanded — buyers are sensitive to price changes.
Q8	c	A negative externality is a harmful side effect on third parties not involved in the market transaction. River pollution harms fishermen who are neither buyers nor sellers of garments.
Q9	b	In the Circular-Flow model, households sell their factors (labour, land, capital) to firms through the Market for Factors of Production, receiving wages, rent, and profit in return.
Q10	b	Marginal thinking (Principle 3) means comparing the extra benefit of one more unit with the extra cost of that unit, rather than looking at totals.

Section B: Short Answer Answers

12 Marks

1. What are Determinants of Demand? List FIVE factors that shift the demand curve. [3]

Definition: Determinants of demand are the factors that cause the entire demand curve to shift. A change in any of these factors changes demand at every price level.

Five determinants (with examples):

1. Consumer Income: Rising income increases demand for normal goods (e.g., more chicken purchased). It decreases demand for inferior goods (e.g., fewer bus rides).

2. Price of Related Goods: If the price of Hilsa fish rises, demand for Rohu fish increases (substitute). If bread prices rise, demand for butter falls (complement).
3. Tastes and Preferences: A doctor recommends more vegetables; vegetable demand increases.
4. Expectations of Future Prices: Before Ramadan, people buy rice early expecting prices to rise. Demand increases today.
5. Number of Buyers: Population growth in Dhaka increases total demand for rice and housing.

2. Explain Market Failure and the role of government. Use a Bangladesh externality example. [3]

Market Failure: Market failure occurs when a free market produces an outcome that is inefficient or harmful for society. In such cases, the market on its own does not allocate resources in the best possible way.

Common causes of market failure:

- Externalities (side effects on third parties not in the transaction)
- Market power (monopoly pricing)
- Public goods (goods the market will not supply adequately)

Bangladesh Example — Negative Externality: Garment factories in Dhaka dump chemical waste into the Buriganga River. This is a negative externality: the factory and its customers are not paying for the damage caused to fishermen and riverside communities. The market, left alone, produces too much pollution. Government intervention — through environmental regulations, fines, or pollution taxes — is justified to correct this market failure. This reflects Principle 7: Governments can sometimes improve market outcomes.

3. Distinguish between Positive Economics and Normative Economics. Give two examples of each. [3]

Positive Economics: Describes the world as it IS. Makes factual claims that can be tested and either confirmed or disproved using evidence and data. It does not involve value judgments.

Example 1: 'When the price of rice rises, quantity demanded falls.' (Testable by data)

Example 2: 'Higher cigarette taxes reduce the number of people who smoke.' (Can be verified empirically)

Normative Economics: Describes how the world SHOULD BE. Involves opinions, values, and ethical judgments about what is desirable. Cannot be proven true or false by data alone.

Example 1: 'The government should subsidize rice for low-income families.' (A value judgment)

Example 2: 'Workers in Bangladesh deserve a higher minimum wage.' (Based on values of fairness)

Key Distinction: An economist acts as a scientist when making positive statements and as a policy adviser when making normative statements.

4. What is Price Elasticity of Demand? Calculate and classify using the given data. [3]

Definition: Price Elasticity of Demand (E_p) measures how sensitive the quantity demanded of a good is to a change in its price.

Formula: $E_p = (dQ / dP) \times (P / Q)$

Calculation ($P_1 = 10$, $Q_1 = 200$ and $P_2 = 8$, $Q_2 = 250$):

$$dP = 10 - 8 = 2$$

$$dQ = 200 - 250 = -50$$

$$E_p = (-50 / 2) \times (10 / 200)$$

$$E_p = -25 \times 0.05 = -1.25$$

$$|E_p| = 1.25$$

Classification: $|E_p| = 1.25 > 1$, therefore demand is ELASTIC.

Interpretation: A 1% increase in price leads to a 1.25% decrease in quantity demanded. Buyers are relatively sensitive to price changes for this good.

Section C: Long Answer

8 Marks

1. Dhaka Vegetable Market — Flood + Population Growth Scenario [8]

(a) Diagram Analysis [2 marks]

Two simultaneous shifts occur in the market:

Supply Shift: The flood in Sylhet destroys crops. This reduces supply. The supply curve shifts to the LEFT (from S_1 to S_2). At every price, fewer vegetables are available.

Demand Shift: The population of Dhaka grows. More buyers enter the market. The demand curve shifts to the RIGHT (from D_1 to D_2). At every price, more vegetables are demanded.

Diagram description (label accordingly):

- Original equilibrium: E_1 at intersection of D_1 and S_1 , at price P_1 and quantity Q_1 .
- S_1 shifts left to S_2 (flood reduces supply).
- D_1 shifts right to D_2 (population growth increases demand).
- New equilibrium: E_2 at intersection of D_2 and S_2 , at a higher price P_2 .
- Quantity at E_2 is ambiguous — depends on the relative size of each shift.

(b) 3-Step Method [2 marks]

Step 1 — Which curves shift? Both curves shift. Supply decreases (flood). Demand increases (population growth).

Step 2 — Direction: Supply curve shifts LEFT. Demand curve shifts RIGHT.

Step 3 — New equilibrium:

Price: Both shifts push the price upward. A decrease in supply raises price; an increase in demand also raises price. Therefore, the equilibrium price DEFINITELY RISES.

Quantity: The two shifts work in opposite directions on quantity. Supply falling tends to reduce quantity. Demand rising tends to increase quantity. The final effect on quantity is **AMBIGUOUS** — it depends on which shift is larger.

- If the population increase is larger: quantity rises.
- If the flood damage is larger: quantity falls.

(c) Price Ceiling Below Equilibrium [2 marks]

Result: A price ceiling set **BELOW** the equilibrium price creates a **SHORTAGE** (excess demand).

Explanation: A price ceiling is a maximum price set by the government. If it is set below the equilibrium price:

- The quantity demanded by buyers (at the lower, controlled price) will be greater than the original equilibrium quantity.
- The quantity supplied by sellers (at the lower price) will be less than the equilibrium quantity.
- Since $Q_d > Q_s$, a shortage results. Buyers compete for a limited supply of vegetables.

In Bangladesh context: If the government sets a maximum price for vegetables below the new market rate, vendors may reduce supply (some may sell in black markets), while consumers try to buy more. Shortages appear on shop shelves. The price ceiling prevents the market from clearing naturally.

(d) Relevant Principle [2 marks]

Principle 7: Governments Can Sometimes Improve Market Outcomes.

Justification: The government intervenes by setting a price ceiling because it wants to protect consumers — especially low-income families in Dhaka — from unaffordably high vegetable prices. This is a form of government intervention intended to improve market outcomes for a vulnerable section of society.

However, Principle 7 also implies that intervention is only beneficial when it corrects a genuine market failure. In this case, the shortage caused by the price ceiling is itself a market distortion created by the policy. A good economist would point out that the intervention, while well-intentioned, may worsen the outcome (shortage) rather than improve it.

Also relevant: Principle 1 (Trade-offs) — The government faces a trade-off between keeping prices affordable and maintaining adequate supply. And Principle 4 (Incentives) — a price ceiling reduces the incentive for suppliers to bring more vegetables to market.

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Tutorial Set: Set 3

Total Marks: **30**

Chapters: 1, 2 & 4

Section A: MCQ Answers

5 Marks

Q	Ans	Explanation
Q1	c	If the price of Rohu fish falls and demand for Hilsa falls too, the two goods move together — they are substitutes. (A fall in Rohu price makes Rohu cheaper; people buy Rohu instead of Hilsa, so Hilsa demand falls.)
Q2	b	Principle 8 states that a country's standard of living depends on its productivity — the amount of goods and services produced per worker per hour.
Q3	c	In a competitive market, no single buyer or seller can influence the price. All participants are price takers — they accept the market price as given.
Q4	b	Economists use the scientific method: observe events, build a theory, then test it using real-world data. Since lab experiments are not possible on whole economies, they rely on natural experiments.
Q5	b	$Q^* = Q_s$ at $P^* = 2$: $Q_s = 10 + 2(2) = 10 + 4 = 14$. Equilibrium quantity is 14.

Section B: Short Answer Answers

10 Marks

1. Define Equilibrium Price and Equilibrium Quantity. Explain what happens when price is above equilibrium. Use the Eid cattle market. **[3]**

Equilibrium Price: The price at which the quantity supplied equals the quantity demanded. It is also called the market-clearing price. At this price, the market is balanced — there is no upward or downward pressure on the price.

Equilibrium Quantity: The quantity of a good bought and sold at the equilibrium price. At this quantity, every willing seller finds a willing buyer.

When price is ABOVE equilibrium — Surplus:

- Quantity supplied exceeds quantity demanded.
- Sellers have unsold goods piling up.
- Sellers respond by lowering prices to attract buyers.
- The price falls back toward equilibrium over time.

Bangladesh Example — Eid Cattle Market: Before Eid-ul-Adha, demand for cattle rises sharply. Supply cannot keep up, causing a shortage and prices rise. After Eid, demand drops suddenly. Many sellers still have unsold cattle — a surplus. Sellers cut prices to clear their stock.

The market naturally moves back toward equilibrium in both cases. No central authority instructs them — prices do the work.

2. Explain the PPF with reference to Scarcity, Efficiency, Trade-offs, and Opportunity Cost. [3]

Production Possibilities Frontier (PPF): A graph showing all the maximum combinations of two goods that an economy can produce with its available resources and technology. The curve bows outward due to increasing opportunity cost.

1. Scarcity: Not every combination of goods is achievable. Points outside the PPF are impossible because resources are limited. The PPF itself is a visual representation of scarcity.

2. Efficiency: Points ON the PPF curve represent efficient production — all resources are fully and productively used. Points INSIDE the PPF are inefficient (resources are wasted, e.g., workers are unemployed).

3. Trade-offs: Moving along the PPF means producing more of one good requires producing less of the other. Society cannot increase both goods simultaneously without new resources or technology.

4. Opportunity Cost: The slope of the PPF at any point represents the opportunity cost of producing one more unit of the good on the horizontal axis. Because the curve bows outward, opportunity cost increases as more of one good is produced — this is called increasing opportunity cost.

3. Explain two reasons why economists sometimes disagree with each other. [2]

Reason 1 — Differences in Scientific Judgments: Economists sometimes have different theories about how the world works, or different estimates of important values. They may agree on the goals but disagree on the facts.

Example: Some economists believe that switching from an income tax to a consumption tax would greatly increase household savings and investment. Other economists believe savings would barely change. Both groups are making different positive claims, leading to different policy advice.

Reason 2 — Differences in Values: Economists may agree on the facts but disagree on what is fair or desirable. These are normative disagreements based on personal, ethical, or political values.

Example: Two economists may both agree that a progressive tax reduces after-tax income inequality. But one may value individual economic freedom more and oppose it, while the other values equality more and support it. The disagreement is about values, not facts.

4. List three determinants of supply (non-price factors). Give a garment industry example for each. [2]

1. Input Prices: The cost of raw materials, labour, and energy used in production. If input prices rise, supply decreases (curve shifts left). Example: If fabric prices rise in Bangladesh, garment factories face higher production costs and reduce output — supply of garments decreases.

2. Technology: Improvements in technology lower the cost of production and allow more output from the same resources. Supply increases (curve shifts right). Example: A garment factory adopts automated stitching machines. Output per worker increases. The factory can supply more shirts at every price level.

3. Number of Sellers: As more firms enter an industry, total market supply increases (curve shifts right). Example: New garment factories open in Narayanganj and Chittagong. The total supply of readymade garments in Bangladesh increases, shifting the market supply curve to the right.

Section C: Application / Analytical Answers

15 Marks

1. Scenario — The Fertilizer Subsidy: Apply THREE of Mankiw's 10 Principles [6]

Principle 4 — People Respond to Incentives: The diesel subsidy is a positive incentive. By lowering the cost of diesel for farmers, the government makes irrigation and mechanized farming cheaper. Farmers respond by using more diesel-powered equipment, irrigating more land, and planting more crops. Output increases. This is exactly how positive incentives change behavior.

Principle 1 — People Face Trade-offs: The government must fund this subsidy from its budget. Every taka spent on the diesel subsidy is a taka not available for education, healthcare, infrastructure, or debt repayment. This is a classic resource trade-off. The government is choosing agricultural support over other public needs. The choice may also involve an efficiency-versus-equality trade-off: the subsidy may be efficient in increasing food output, but it may disproportionately benefit larger farmers who use more diesel.

Principle 7 — Governments Can Sometimes Improve Market Outcomes: If the free market alone is not producing enough food — perhaps because small farmers cannot afford diesel at market prices — this is a market failure. The subsidy is a justified government intervention to correct this failure by making food production more affordable and increasing agricultural output. This benefits the wider economy through lower food prices.

Principle 8 — A Country's Standard of Living Depends on Its Ability to Produce: If farmers produce more rice, wheat, and vegetables, the country's overall productivity rises. Higher agricultural productivity means more food at lower prices, improving the standard of living — especially for lower-income households who spend a large share of income on food.

Conclusion: The subsidy has positive effects on food production and farm income, but it carries an opportunity cost to the government budget and may create long-term dependency if not designed carefully.

2. Numerical Problem — Ice Cream Market: $Q_d = 30 - 5P$ and $Q_s = -5 + 5P$ [6]

(a) Find P^* and Q^* [2 marks]

Set $Q_d = Q_s$:

$$30 - 5P = -5 + 5P$$

$$30 + 5 = 5P + 5P$$

$$35 = 10P$$

$$P^* = 3.5 \text{ (Equilibrium Price)}$$

Find Q^* using demand equation:

$$Q^* = 30 - 5(3.5) = 30 - 17.5 = 12.5$$

$$Q^* = 12.5 \text{ units (Equilibrium Quantity)}$$

Verification using supply equation:

$$Q_s = -5 + 5(3.5) = -5 + 17.5 = 12.5 \text{ (confirmed)}$$

(b) Surplus or Shortage at $P = 4$ [2 marks]

At $P = 4$:

$$Q_d = 30 - 5(4) = 30 - 20 = 10$$

$$Q_s = -5 + 5(4) = -5 + 20 = 15$$

Since $Q_s (15) > Q_d (10)$: There is a SURPLUS of 5 units.

Reason: $P = 4$ is above the equilibrium price of 3.5. At this higher price, sellers want to supply more than buyers are willing to purchase. The market has unsold stock. Sellers will lower the price back toward $P^* = 3.5$.

(c) Sugar Price Rises — Supply Shifts Left (Diagram Description) [2 marks]

Sugar is an input in ice cream production. When the price of sugar rises:

Step 1: The supply curve is affected. Higher input cost reduces profitability. The supply of ice cream decreases.

Step 2: The supply curve shifts to the LEFT (from S_1 to S_2). At every price, sellers supply fewer cones.

Step 3: The demand curve is unchanged. At the old equilibrium price of $P^* = 3.5$, quantity demanded (12.5) now exceeds the reduced quantity supplied — a shortage exists. Sellers raise prices.

New equilibrium (diagram outcome):

- Equilibrium Price RISES (above 3.5).
- Equilibrium Quantity FALLS (below 12.5).

Diagram: Draw original S_1 and D crossing at E_1 ($P^* = 3.5$, $Q^* = 12.5$). Draw S_2 to the left of S_1 . New equilibrium E_2 is at a higher price and lower quantity.

3. Bangladesh Bank's Dilemma — Principle 10 and the Phillips Curve [3]

Principle 10: Society faces a short-run trade-off between inflation and unemployment. When one falls, the other tends to rise. This inverse relationship is shown by the Phillips Curve.

The Dilemma: Bangladesh Bank currently faces both high inflation (9%) and high unemployment (7%). These two problems call for opposite policy responses, making it impossible to solve both simultaneously in the short run.

Option A — Fight Inflation (Tighten Monetary Policy):

- Bangladesh Bank raises the interest rate.
- Borrowing becomes more expensive. Businesses invest less and hire fewer workers.
- Unemployment rises further (worsens the existing problem).
- But the higher rate reduces money in circulation, slowing inflation.
- Outcome: Inflation falls, unemployment rises.

Option B — Reduce Unemployment (Ease Monetary Policy):

- Bangladesh Bank lowers the interest rate.
- Cheaper credit encourages businesses to borrow, invest, and hire.
- Unemployment falls.
- But more money in the economy pushes prices higher.
- Outcome: Unemployment falls, inflation worsens.

Standard Policy Advice: Most economists advise controlling inflation first. High inflation functions as a hidden tax on savings and disproportionately harms low-income households (who cannot protect their savings). Once inflation is controlled, the bank can gradually ease policy to reduce unemployment.

Long-run solution: The trade-off exists only in the short run. In the long run, investments in education, skills training, and technology (improving productivity) can reduce both inflation and unemployment simultaneously — escaping the trade-off entirely.